

DISCRETE MATHEMATICS -411 ASSIGNMENT NO.7

CHAPTER 7

EXERCISE 7.1

1. What is the probability that a card selected at random from a standard deck of 52 cards is an ace?

3. What is the probability that a randomly selected integer chosen from the first 100 positive integers is odd?

13. What is the probability that a five-card poker hand contains at least one ace?

21. What is the probability that a fair die never comes up an even number when it is rolled six times?

27. Find the probability of selecting exactly one of the correct six integers in a lottery, where the order in which these integers are selected does not matter, from the positive integers not exceeding
c) 56. d) 64.

29. In a superlottery, players win a fortune if they choose the eight numbers selected by a computer from the positive integers not exceeding 100. What is the probability that a player wins this superlottery?

33. What is the probability that Abby, Barry, and Sylvia win the first, second, and third prizes, respectively, in a drawing if 200 people enter a contest and

a) no one can win more than one prize.

b) winning more than one prize is allowed.

EXERCISE 7.2

1. What probability should be assigned to the outcome of heads when a biased coin is tossed, if heads is three times as likely to come up as tails? What probability should be assigned to the outcome of tails?

5. A pair of dice is loaded. The probability that a 4 appears on the first die is $\frac{2}{7}$, and the probability that a 3 appears on the second die is $\frac{2}{7}$. Other outcomes for each

die appear with probability $\frac{1}{7}$. What is the probability of 7 appearing as the sum of the numbers when the two dice are rolled?

23. What is the conditional probability that exactly four heads appear when a fair coin is flipped five times, given that the first flip came up heads?

27. Let E and F be the events that a family of n children has children of both sexes and has at most one boy, respectively.

Are E and F independent if

a) $n = 2$? b) $n = 4$? c) $n = 5$?

EXERCISE 7.3

1. Suppose that E and F are events in a sample space and $p(E) = \frac{1}{3}$, $p(F) = \frac{1}{2}$, and $p(E \mid F) = \frac{2}{5}$. Find $p(F \mid E)$.

3. Suppose that Frida selects a ball by first picking one of two boxes at random and then selecting a ball from this box at random. The first box contains two white balls and three blue balls, and the second box contains four white balls and one blue ball. What is the probability that Frida picked a ball from the first box if she has selected a blue ball?

EXERCISE 7.4

1. What is the expected number of heads that come up when a fair coin is flipped five times?

7. The final exam of a discrete mathematics course consists of 50 true/false questions, each worth two points and 25 multiple-choice questions, each worth four points. The probability that Linda answers true/false question correctly is 0.9, and the probability that she answers a multiple-choice question correctly is 0.8. What is her expected score on the final?